

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE - 4

COMPOSITION OF ATMOSPHERE

- **PARTICULATE MATTER (PART-3)**

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

FORCING: THE INITIAL DRIVERS OF CLIMATE

FORCING IN “LAYMAN” TERMS

The term "forcing" means a change which may "push" the climate system in the direction of warming or cooling. Solar irradiance, GHG emissions, Particulate matter are some of the examples of forcing.



SOLAR IRRADIANCE

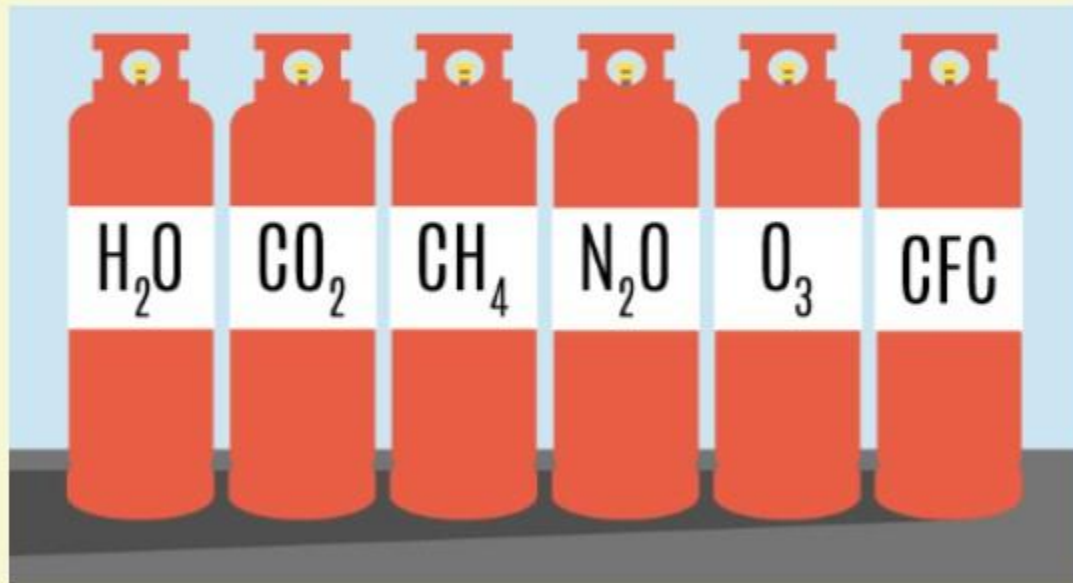
- 1 Solar heat is the important source of heat for the Earth.
- 2 The variance in solar radiation is a form of forcing. One such variation is sun spot cycle.
- 3 The sun has a 11-year sun spot cycle, which causes about 0.1% of the variation in the sun's output.

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

FORCING: THE INITIAL DRIVERS OF CLIMATE

GHG EMISSIONS



- 1 Since the industrial revolution, concentrations of greenhouse gases such as carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O) have risen in the atmosphere.
- 2 These greenhouse gases absorb and then re-radiate heat in Earth's atmosphere which causes increased warming.

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

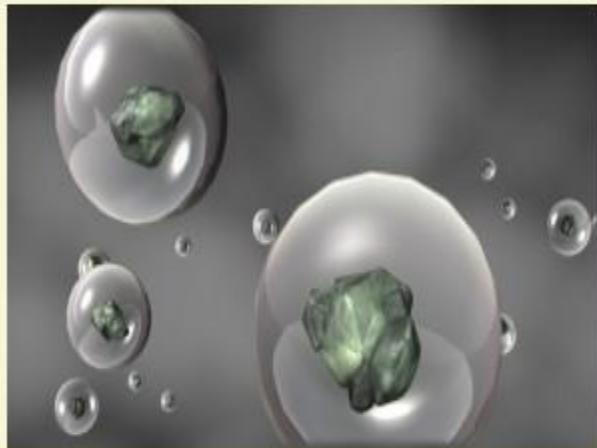
FORCING: THE INITIAL DRIVERS OF CLIMATE

AEROSOLS AND PM

Some aerosols absorb heat while others reflect them back and by this way contribute to climate change.

HOW IS AEROSOL DIFFERENT FROM PARTICULATE MATTER?

- Aerosols are suspensions of solid or liquid particles in a gas (usually air).
- The particulate portion of an aerosol is referred to as Particulate Matter, or PM.
- Particulate Matter is a generic term applied to chemically heterogeneous discrete liquid droplets or solid particles.

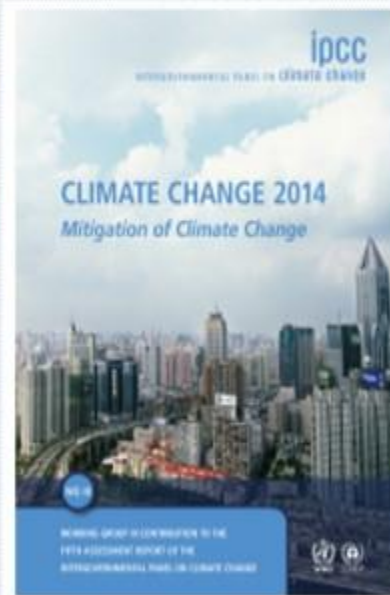


CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

RADIATIVE FORCING

FIFTH ASSESSMENT REPORT OF IPCC- THE GAMECHANGER!!

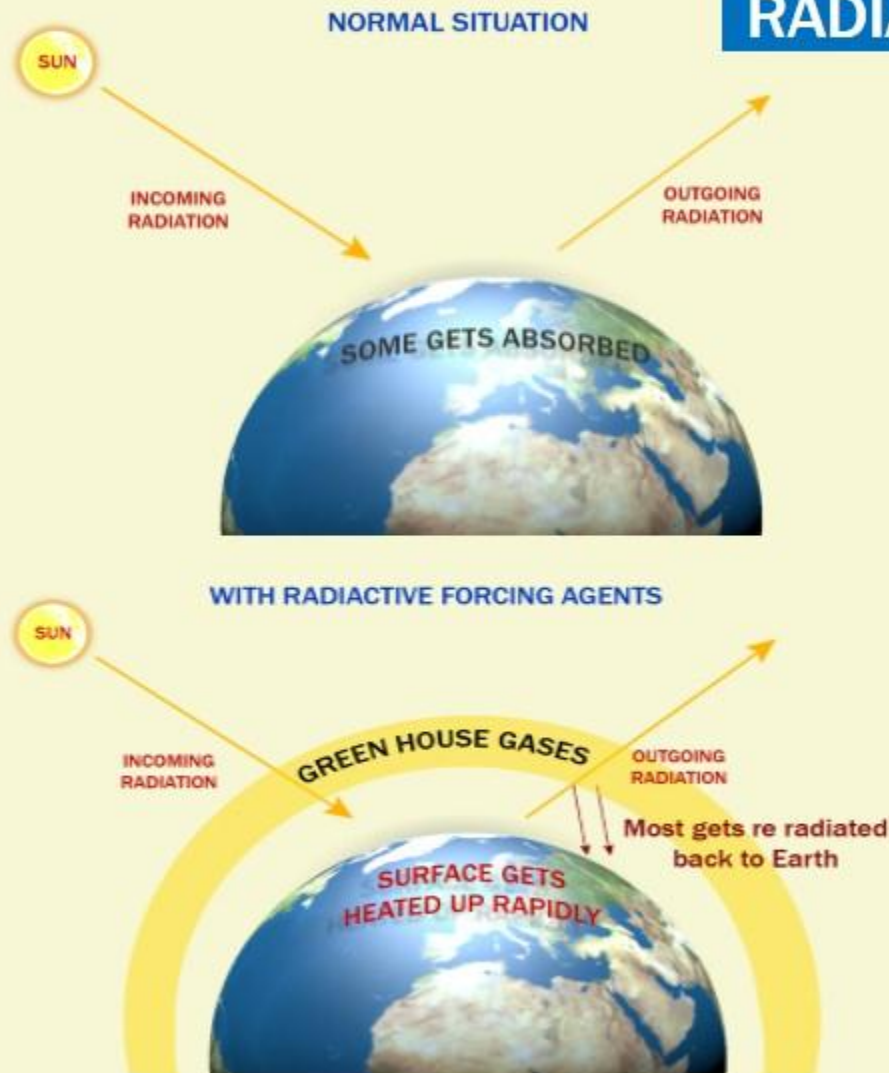


- 1 The Fifth Assessment Report (AR5) of the Intergovernmental Panel on Climate Change (IPCC) first identified the impact of forcing on climate change.
- 2 It gave the term of “**effective radiative forcing**” that redefined the global outlook towards climate change.

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

RADIATIVE FORCING



- 1 The Earth's surface temperature is determined by the **balance between incoming solar radiation and outgoing infrared radiation**.
- 2 Radiative forcing or climate forcing is the **difference between insolation (sunlight) absorbed by the Earth and energy radiated back to space**. It is the **capacity of a gas or other forcing agents to affect that energy balance**, thereby contributing to climate change
- 3 It is expressed in Watts per square meter (W/m^2)

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

RADIATIVE FORCING

NORMAL SITUATION



WITH RADIATIVE FORCING AGENTS



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PARTICULATE MATTER

RADIATIVE FORCING

RESIDENCE TIME OF GHG

Greenhouse gas	Average lifetime in the atmosphere
Carbon dioxide	50-200 years*
Methane	12 years
Nitrous oxide	120 years
CFC-12	100 years
CFC-11	45 years
HFC-134a	14.6 years
Sulfur hexafluoride	3,200 years

FACTORS THAT DETERMINE CAPACITY OF RADIATIVE FORCING AGENT

- 1 Amount/Atmospheric Concentration
- 2 Warming or Cooling Capacity
- 3 Spatial Distribution
- 4 Duration/Residence Time in the Atmosphere

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

POSSIBLE MAINS QUESTION

TOPIC:

GS Paper I : Important Geophysical phenomena
-earthquakes, tsunami, volcanoes, cyclones

GS Paper III : Environmental pollution and degradation

- 1** How does particulate matter contribute to climate change?
(12 Marks)(250 words)
- 2** How can volcanoes bring about climate change? Explain with the help of past geographical examples. (10 marks)(200 words)

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

PARTICULATE MATTER REINFORCES CLIMATE CHANGE

BLACK CARBON/SOOT



SULPHATE AEROSOLS

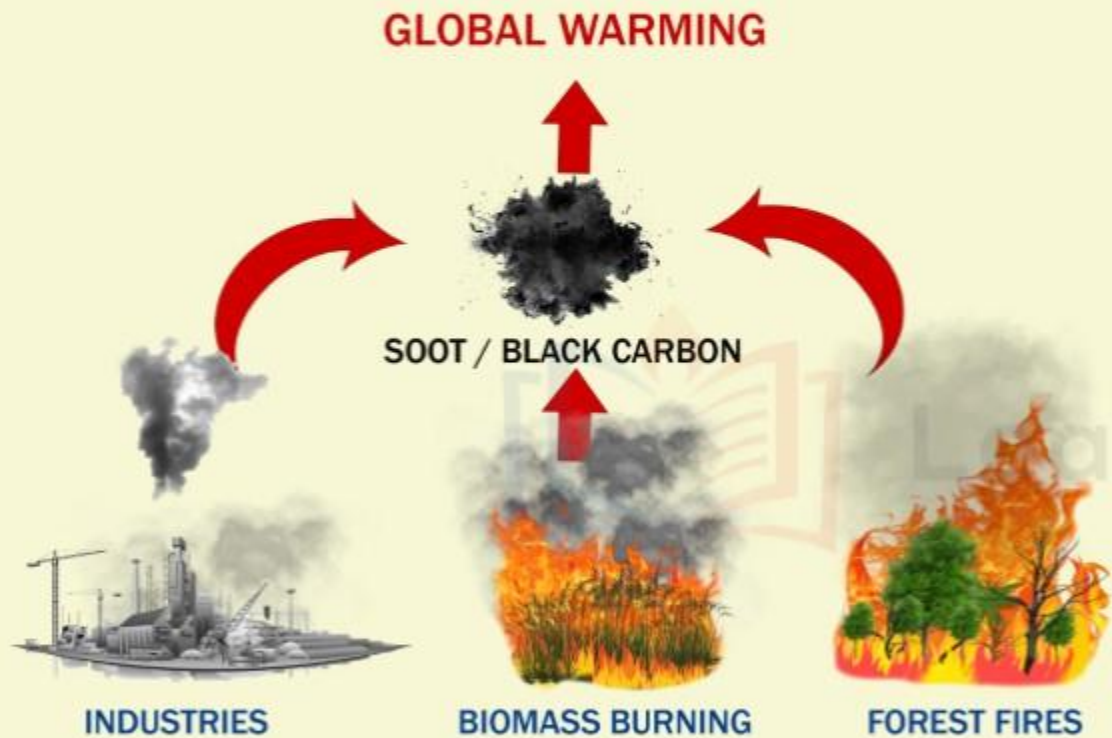


- 1 PM affect the climate of the earth by changing the amount of incoming solar radiation and outgoing terrestrial long wave radiation retained in the earth's system.
- 2 PM like Black Carbon warms the earth by more absorption of heat.
- 3 Whereas sulphate aerosols cool down the earth.

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

RADIATIVE FORCING



POSITIVE RADIATIVE FORCING

- 1 Positive radiative forcing means that the Earth receives more incoming energy from sunlight than what it radiates back to space and ultimately leads to warming.
- 2 Aerosols like Black Carbon absorb the radiation and warm the Earth.

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

RADIATIVE FORCING

QUICK FACTS FOR EXAMINATION!



BLACK CARBON

- CO_2 has a longer lifetime therefore reducing it becomes difficult.
- On the other hand BC has a short lifetime (only for a few weeks). So cutting its emissions could go a long way to slowing climate change particularly in the rapidly changing Arctic.

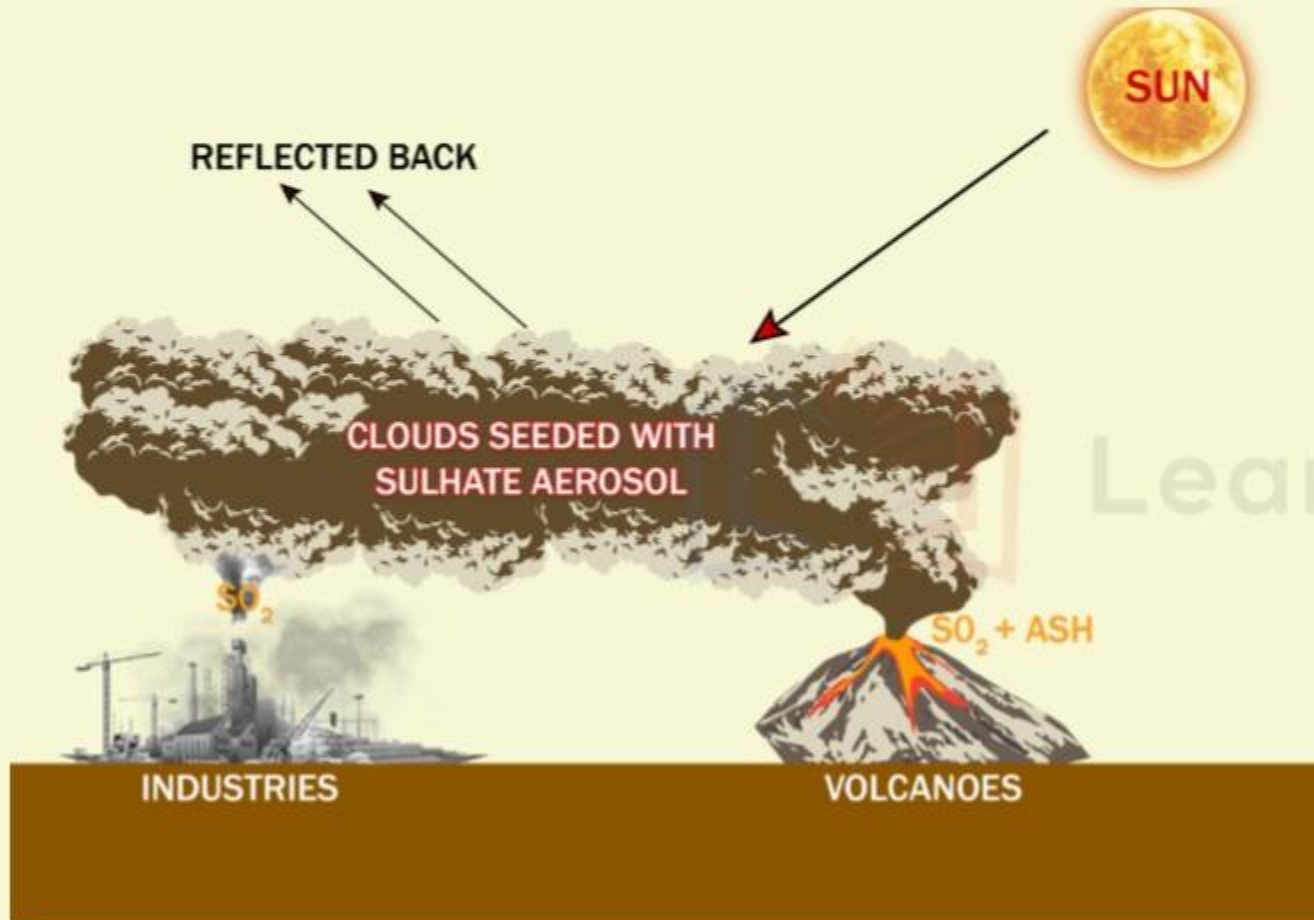
WHAT DOES BC DO TO POLAR ICE?

- It affects the albedo of the place.
- Bright surfaces reflect radiation and cool the climate, whereas darker surfaces absorb radiation and produce a warming effect.
- In the Arctic, especially BC from wildfires and industrial pollution are hastening the melting of ice.

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

RADIATIVE FORCING



NEGATIVE RADIATIVE FORCING

- 1** Negative radiative forcing means Earth loses more energy to space than it receives from the sun and ultimately leads to cooling.
- 2** Sulfate aerosols, which result from burning coal, biomass, and volcanic eruptions, tend to cool the Earth.

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

RADIATIVE FORCING

NEGATIVE RADIATIVE FORCING

HOW VOLCANOES CAN COOL DOWN EARTH?



PAST PRECEDENTS FROM NATURE

YEAR	VOLCANO	PLACE	REMARKS
1963	Mt.Agung	Indonesia	Temp dipped by 0.2°C for 2 years
1991	Mt Pinatubo	Philippines	Temp dipped by 0.6°C for 2 years as it injected 20 megatonnes of SO_2

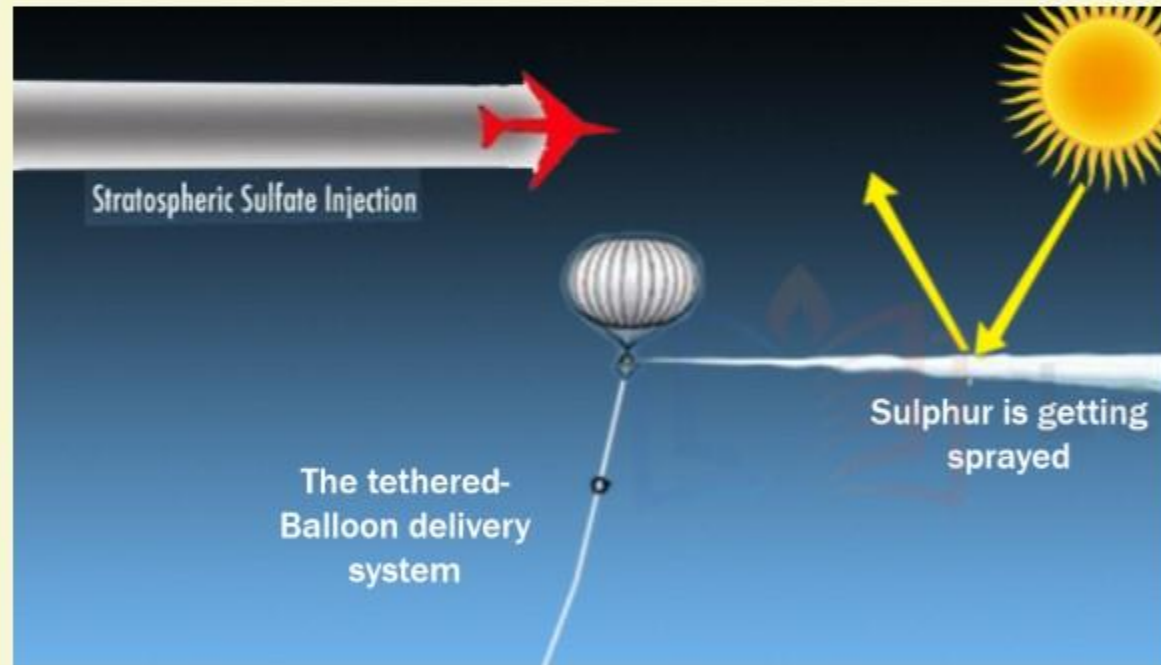
- 1 In the short term, dust particles would cause regional cooling, as the layer of dust prevents some sunlight from reaching the ground.
- 2 In the long term, Sulphur dioxide would mix with water droplets in the atmosphere, spread across the globe and reflect sunlight for up to three years because of which average global temperature could decrease significantly.

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

RADIATIVE FORCING

“CLIMATE ENGINEERING” IS THE BUZZWORD



STRATOSPHERIC AEROSOL INJECTION

- 1 SAI models the natural volcano to cool the Earth.
- 2 SAI is the process of spraying into the stratosphere fine, light-colored particles designed to reflect back part of the solar radiation before it reaches and warms the Earth

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

RADIATIVE FORCING

“CLIMATE ENGINEERING” IS THE BUZZWORD



IS SAI THE PANACEA TO CLIMATE CHANGE?

- 1 Injection of Sulphur compounds into the stratosphere is **likely to increase acid deposition** on the ground.
- 2 Also when the injection is **discontinued**, there can be rapid re-warming.
- 3 There is a potential for the disastrous side effects **crossing national boundaries**.

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

FEEDBACK: AMPLIFY OR DIMINSH FORCINGS

Climate feedbacks may positive (+ve) and negative (-ve)

 Met Office Climate feedbacks can be positive and negative



1 The forcings "feed back" into the climate system and change the amount of warming.

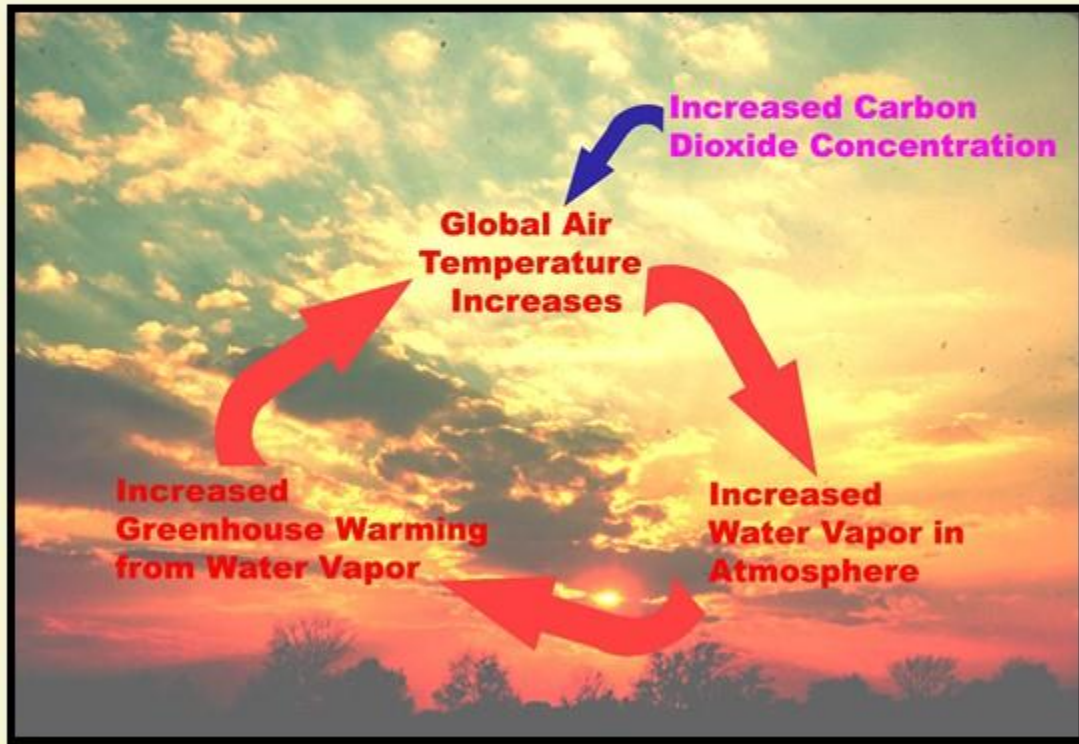
2 These sorts of loops whereby the warming of the Earth and the elements that cause warming (or cooling) interact with one another are called feedbacks.

3 A climate feedback is a secondary change— either positive or negative — that occurs within the climate system in response to a forcing mechanism.

CONSTITUENTS OF ATMOSPHERE

PREVIOUS DISCUSSION IN MODULE 1

2. WATER VAPOUR

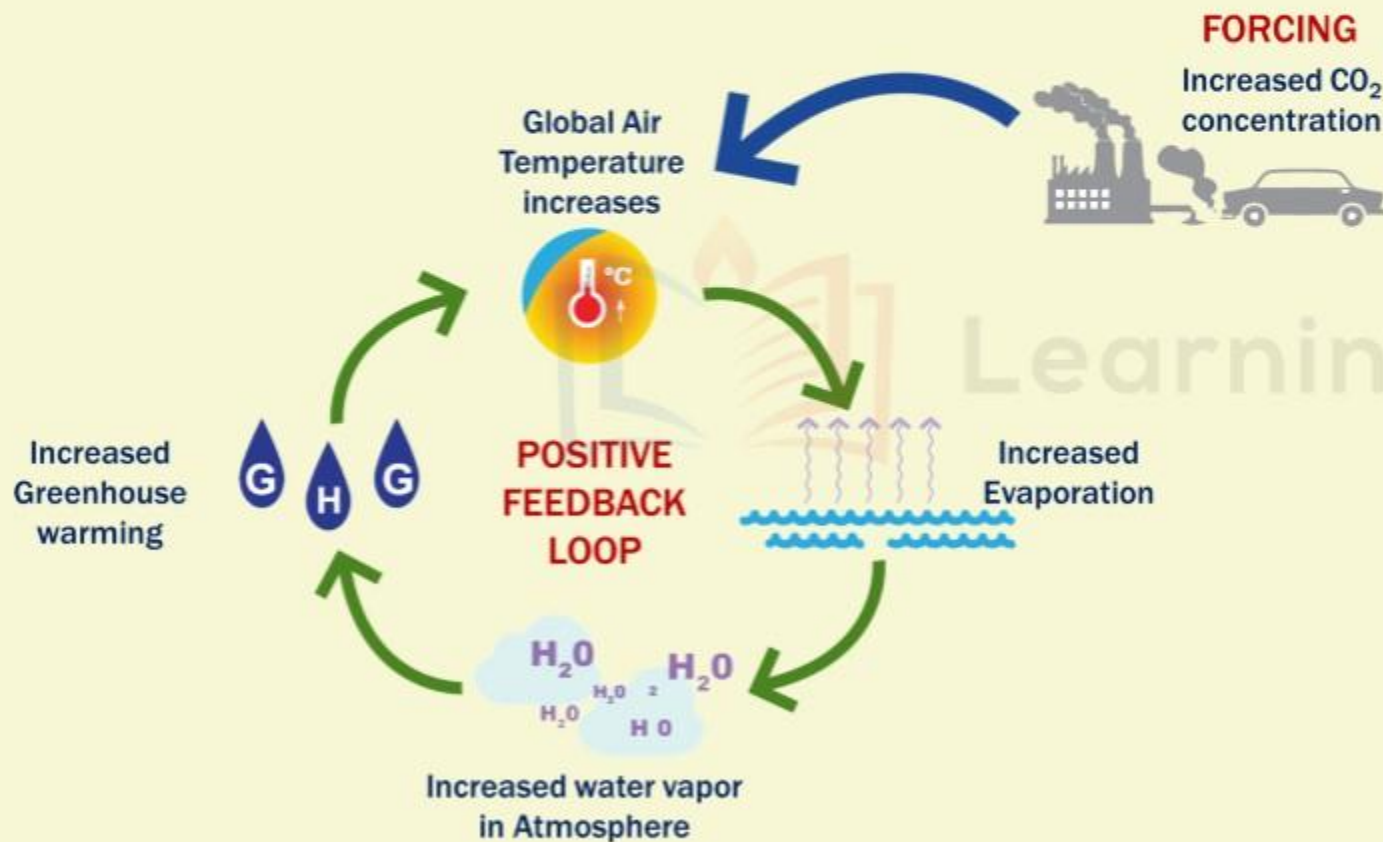


- More than 90% of the total atmospheric vapour is found upto the height of 5km.
- Vapour depends on temperature.
- More the evaporation (as in tropical areas) more will be the vapour content.
- Responsible for trapping the outgoing long-wave radiation.

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

FEEDBACK: AMPLIFY OR DIMINSH FORCINGS



POSITIVE FEEDBACK LOOP

The main positive feedback in global warming is the tendency of warming to increase the amount of water vapor in the atmosphere, which in turn leads to further warming.

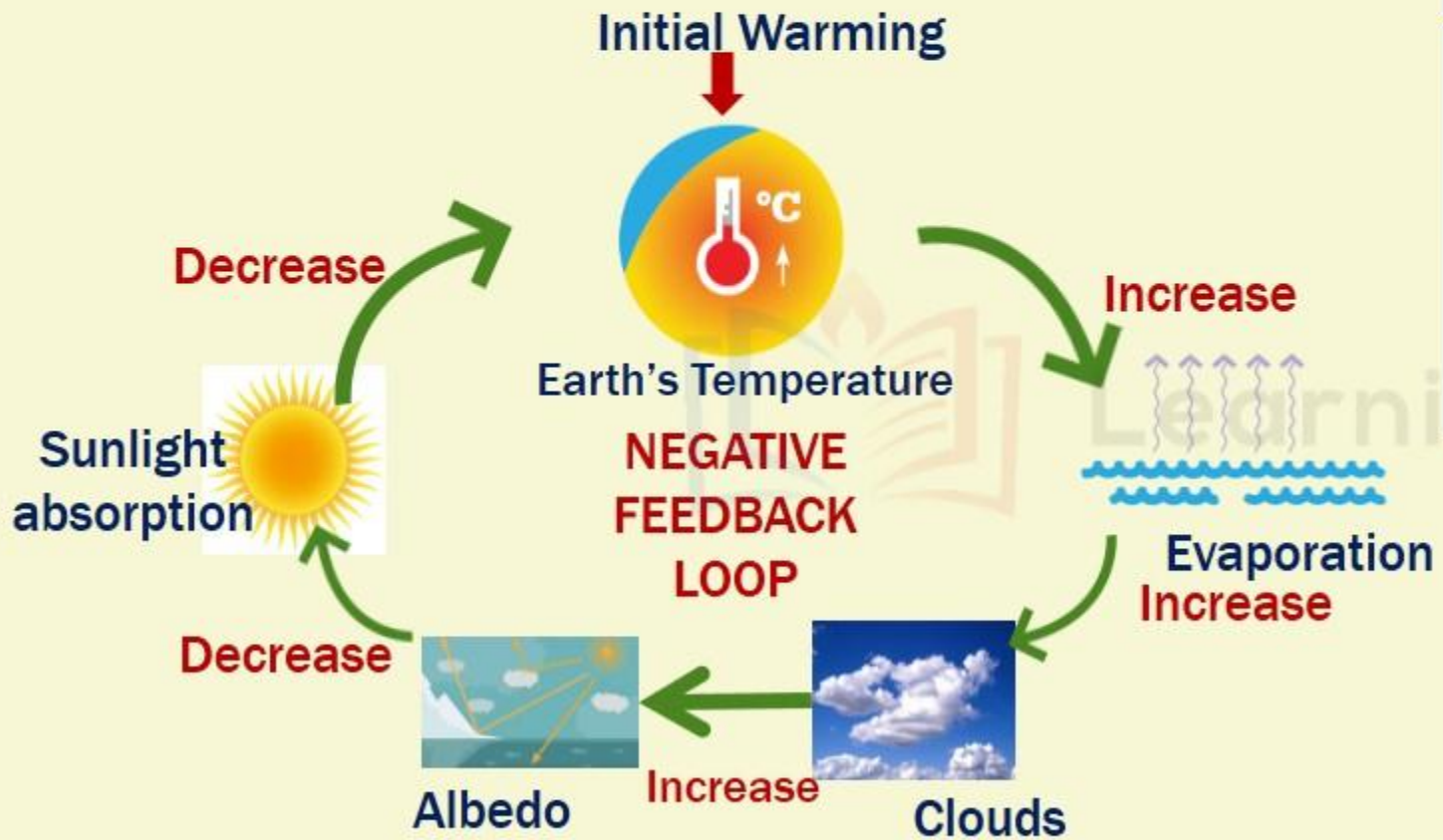
CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

FEEDBACK: AMPLIFY OR DIMINSH FORCINGS

NEGATIVE FEEDBACK LOOP

The main negative feedback comes from the Stefan-Boltzmann law, the amount of heat radiated from the Earth into space changes with the fourth power of the temperature of Earth's surface and atmosphere.



Stefan-Boltzmann law

$$P \propto T^4$$

Power
Radiated (Watts)

Temperature
(Kelvins)

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suggestions@learningspace.in

To Contact us:

info@learningspace.in

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